

The first lecture is on orientation. What am I trying to do? The purpose of this course is to prepare you for your technical future. There really isn't this course any technical content, although I'm going to talk about digital fillers and all kinds of things. There are things you presumably know. I am concerned about style.

I have studied great scientists, ever since I was at Los Alamos during the war. What is different between those who do and those who do not do significant things? Mainly, it's a manner of style.

Many a person I've known worked just as hard and others, but didn't have much to show for it. So my problem is, to instill in you something called style so you'll amount to something.

After all, the Navy is paying a large sum of money to have you here. And it wants its money back, by your later performance.

Now I will examine, criticize and talk about various people's style. Mainly my own, but other people's, why can we use it. Now, there are many things I'm going to tell you, I wish somebody had told me. I had to find out for myself. This course is not a normal technical course. It's all about the topics they never told you in class, but they should have. Because each course is taught this way and a large amount falls in between. That's why I'm trying to pick up.

Now style cannot be put into words. I can only approach you by particular examples and let you infer what it is. Now, there's a belief that you probably have, that anything can be talked about. This goes back to Socrates, Plato, Aristotle and the early Greek times. They thought they could talk about the gods, truth, beauty, justice, love, all those things. At the time they were saying these things, there were the mystery cults in Greece. Who said you must experience, you cannot talk.

And if you remember the Middle Ages, various saints said, you can't talk about God. You got to experience him. The same way the Mohammedans about Allah, you can't portray him, you can't put pictures. You must sense. So there is a long minuscule that says you cannot put everything into words. And one of them is style. I really cannot say what I mean, I can only give you these examples of struggle. Hopefully, you will get the idea.

Now to be effective at a course like this, I have found that I have to talk about myself. If I make abstract remarks. It just sounds like so many pious words. If I talk about me and what I've done, maybe it will penetrate you. Now it gives a course, attitude of bragging. I'm always talking about myself. But I will tell you several mistakes that I made, so you won't do the same sort of thing.

Similarly, I have to get you to quit your modesty. I have to get you individually, to respond to my challenge that you're going to be great. You have to say to yourself, "Yes, if that guy Hammond can go out and become a great scientist, I can. Or I can become a great person." I have to get you to say to yourself that you want to. That's it's worth the effort. And you're going to try to be something more than just the average person.

Now, while we speak of teachers, we are really coaches. I cannot run a mile, the four minute mile for you. I can comment upon your style. But you know you must do the work. The same way I cannot make you a great scientist. I can criticize style and other things. But I cannot, by mere words make you a great scientist. You just as in running four minute mile, must do the work. Which means you have to take what you hear and read. Think it over carefully, discuss with your friends and see what you can adapt yourself.

There is no one style which is successful. Painters paint many different styles. You have to find a style that fits you. Which means you have to take what fragments you can from other

people, use them and adapt them and become yours. You can't copy me directly, you won't get away with it. And I will use the analogy of painting as an example. In painting, once you've learned color mixing and form and sketching, and so on, you study under a master who you temporarily accept as knowing what he's talking about.

Well, there are limits what can be done. You know that you copy the master style, exactly, you will not be a great painter. You know also, that if you paint in the style he did, or she did, it's too late. The future wants a different style. Thus I can tell you about the style I used in the past. But that won't be the style you'll have to cope with the future. You must manufacture the style, which will make you as a significant person in the future. So it's not easy. While I can only talk about past ones and make references to possible future ones. It's a problem you face. What I did would not make me successful if I were starting now. Just as my predecessor got successful on other things that I couldn't do and get successful on.

Now is another practice very difficult for you. When I went to build our laboratories 1946, I looked around since I was already interested what made great scientists. And I looked at what they did. And when I looked at what they did to become famous, it didn't look that difficult. They tend to do the easy problems. Now I found in the course my time there, a couple of holes they left. But fundamentally, they did the easy problems. My generation did somewhat harder ones, and we left to the others the harder still. Every generation has more difficulty, but you stand on our shoulders to some extent yet, the task is harder. Having got man to the moon, the next real good feat in space is gonna be a lot harder. Therefore you have difficulty, it's very definite.

Now when I came to Bell Labs, there were four of us at the same time about. We came in about the same time. And we were about the same age within a year. We probably called

ourselves the Four Young Turks. And many, many years later, I discovered top management called the same. We were troublemakers. We didn't do things the way the previous generation did. We did new things.

The previous generation didn't like it. We didn't do things right. For example, my boss Henry Boda in network theory had made reputation doing network to with complex variable, and knew that's how you do things, after all. That is what made him famous. This guy Hamming comes on and keeps using computing machines, which is not the way to do it in his eyes. But it was the thing that needed to be done. This is a lesson which I want to get across to you regularly.

Supposing I am successful and you do rise to the top. Would you please remember that what made you great is not appropriate for the next generation. You know how to get great because after all, you were great. But the things that you did may not be appropriate for next generation. All too often we have a troubled bosses. They know by God, this is the way I did it and I got the top, then it must be right. They're very often wrong. And I want you to think seriously, when you rise to the top that your method of success is not appropriate. Now the world has changed.

I want to talk to education. Education is what, when and why to do things. Training is how to do it. Most year courses I've been training, I'm trying to talk about the education part. It's not easy. But the school has allowed me a great deal of latitude in putting this course together, which is concentrating on education. Now, if you have one without the other, it's not much good. I've had very able technical people reporting to me, who apply their technology and the methods to the wrong problem. And it had to be undone. I have other people, who had all kinds of theory but couldn't do anything. They're not what you use

either. You need both theory to guide you and skill and technique to do. One without the other isn't too good.

Now, in a certain sense, I'm engaged in meta education. I'm talking about education constantly, because that's what you're going to have to do. You're going to have to educate yourself constantly. That's what the future says. Now I'm going to constantly try and project forward what the world's gonna be like.

Let's look back first history. The modern era in science engineering began with Sir Isaac Newton, roughly. 1642, he was born Christmas Day, the same year that Galileo died. And he lived to be about 85. So we can say it's around 1700. From Newton's time to ours, we have about double the knowledge every 17 years.

The doubling period of science from then to now is roughly 17. Why can't a Bell Laboratories in 46, they were trying to shrink down the war size down to 5500 people. I watched through 30 years of management, putting hiring freeze and doing everything else like that double every 17 years with small Wiggles. They had to hire two people to keep up with expanding knowledge. Publications, books, journals, and so on. For example, I think I have the numbers here. No, I guess I don't. I'm going to make a digression. Oh. The other thing about the situation is that 90% of the scientists who ever lived are now alive. It's a common statement. I'm going to now turn to a back of the envelope calculation, which I learned by watching family and other people, and other Shockley people I used to get lunch with them. I'm going to suppose first we have an exponential growth of the number of scientists. That comes from a differential equation, the rate of change is proportional how much you have. And the solution is a you know, the exponential growth. Now if I assume that the amount of knowledge being generated is proportional to the number of scientists, this is the amount of

rate and in the up to 17 years ago. This is how much we generated. This is about up to now.

Now I put minus infinity on because it doesn't matter what lower limit I put is so small, doesn't matter. The exponential is very, very small. So who cares?

Well, I will be would simply work it out. I would do the integration. I come up that, and the statement was half the stuff has been done, the doubling every 17 years from 17 years ago, now we've doubled. That says the ratio to half. I've got a formula for B. Now take the other statement, 90% of the scientists who ever lived are now alive. From now back 55 years, that's what I'm going to take for lifetime of a scientist. You probably don't mean a living scientist, when he's two years old, you probably mean his science alive, what he's become are beginning to be a scientist. And until he decay somewhere in the 80s you deserve a sign. So 55 years is a reasonable number. If I put that in over the whole, of all scientists that ever lived, I come up with this, using that, this is a B. I come up with .9 which is just close enough to 90%.

Now let's see what happened. I got a clear idea what I was talking about. And I had to answer the question which I hadn't thought about. What did I mean by a lifetime of a scientist? But you see, those two statements are compatible. We double every 17 years, and 90% of the scientists who ever lived are now alive. You have seen enormous growth of science from Newton's time to now. Well, let me project. Well, let me say now, a good estimate of the number of various branches of science which we have developed. In Newton's this time, we had only one thing called natural philosophy. Now we have lots of specialties. There are something like 10,000 specialties. There certainly is more than 1,000, and almost certainly less than 100,000. So 10,000 is a good number.

Now if I could check forward, doubling every 17 years, for 340 years, that's a million fold to the 20th. That would make 10 million fields especially. But you don't believe it. You don't believe in 340 years, that'll be 10 billion fields of specialty. Consequently, science cannot go the way it has been for the next 320 or 40 years. The doubling and the growth cannot go on. One of the things we have done is we've got an exponential number of people in the field. We can't go on that either. Everyone would have to be a scientist. So you know the past is not too good a guide to the future.

Now the reason why I want to put the back the envelope in is it's widely used. I observed that Fairmi and Shockley and those, I use to eat lunch with them. They did back the envelope. And you saw what I had to do. Not only that, but it also does two things. It puts the thing firmer in your mind, having shown you the calculation, you may retain a little longer. Plus it gives you practice in quick modeling. Nobody pretends this is really accurate. I don't pretend 17 is exactly a number. It's somewhere around there. But back the envelope calculations are very useful.

I find it very, very useful. When I hear things over TV or something else. Radio, read newspapers, so on to a quick modeling and ask myself, are these numbers possible? And very frequently, two things emerge, either they're not possible or B, you didn't even know what they were talking about to make a model. Your father, they failed to tell you what they were talking about, just gave you a spectacular answer. So doing back envelope modeling is a very, very big help.

Now, this doubling business is a very serious one. I've had lived through my life with that fact. So I put them in here a table. Double the 17 years, triple that four, five, six, seven, eight, nine, ten times, about 56 years, something like that. Hey, how do you read that? One way is

ask the time from now to retirement. Look at this column. That's how much knowledge will be, that much times at what you now have. If we go on the same way. You face a rather horrendous future. Another way to look at is this.

Suppose you were 34 when your child was born. Now your child goes to college. There's four times as much knowledge, not just mathematical theorems. Recording is Beethoven's Ninth, where to go skiing, what channels to read, listen to on TV. There's going to be four times as much knowledge for your poor child to face. Now you remember when you hit college, how much there seemed to be? Don't be surprised if your children are somewhat more disoriented than you were. And God knows you were sometimes disoriented.

This is what that means. Furthermore, the doubling, all the doubling occurs worst in the last period. Almost half, the half the episodes occur in the last doubling period. And that's what causes saturation. Saturation comes on quite rapidly. So another way of looking at doubling is simply this table here, which is disconcerting. If you think you'll be chief of staff in, say 44 years, no, I'll say 39 years. They'll be five times as much knowledge needed to run the Navy, as needed now.

That is what you face. But what's my answer? My answer to that is learning to learn, was the only thing I could do. Things become obsolete. Something like half of what we have taught you. Loving the other courses will be obsolete in 15 years. Either we're no longer doing it, or it's been replaced by something else. Consider what I had lived through. I came to Bell Laboratories in 46, and they were running back and due to so on with a very important part. So I started having a mathematical background, studying light to engineering and what back and tubes were to song.

But in some years, I began eating with the physics department and I ate with the guy is while they were perfecting that when they started. When they were developing engineering side of transistors, I did a great deal of calculation for him on transistors. I absolutely needed all the knowledge I knew. I have to to see if I could do for long walks up to my friends office where he keeps going around the show suits what a vacuum tube is, you don't see the very often. Now you can say well, the original transistor roll tin cans and three legs. Now there's a minimalist ship that sides. I've had to do that. At Los Alamos, we calculate how we bomb designs, on really calculus, which probably averaged maybe an operation or a second, or maybe a second and a half. Round the clock, six and a half days a week for a month. Sometimes three months, but typically about a month, to get one solution. Now you can punch in a modern machine, go boop, boop, and there's the answer. I've had to live through a tremendous change.

Furthermore, I was educated as a mathematician, I certainly had no course numerical analysis. I never knew about a computer. I knew a little physics, the Los Alamos taught me so more. But fundamentally, when I went to Bell Labs, because I believe that the computing I did, I should understand the nature problem. I had to learn something of the breadth of physical sciences. Some chemistry, as well a lot of physics, some social science and a little bit of biological science. Because laboratories had such departments, and sub social science. I spent a lifetime getting background knowledge on something, you have to have background knowledge enough to penetrate jargon, which I'll talk about extensively at a later date.

Now, one thing you could do is to try and claim the fundamentals, which is very glib until you ask what do I mean by fundamentals? Well, I have two criteria, which are not adequate. One is from the fundamental you can derive the rest of the field. Secondly, they've been around

for some time. But the fundamentals of application, which were vacuum tubes, doesn't count now. True, the formula for gain, oh. I have trouble with names frequently. I'll come to it pretty soon. Nyquist formulas are still good. The game form is used out of back in tubes are still useful. Although we have to apply it to other things. Feedback is still the same. A lot of things are not.

Now I need to discuss science versus engineering. Science, if you are doing it, you shouldn't know what you're doing. If you know what you're doing, you shouldn't be doing it. Not in science, because science is supposed to be exploration. But you don't know. Engineering. You shouldn't be doing it unless you do know what you're doing. Well, nothing is pure. Science involves a great deal of engineering. And engineering involves a great deal of new material. So it's a great blend. But what is painful to you is going to be worse, is that they two fields are growing together because of a simple fact. Again, going back by first candy Bell Labs, when suddenly we discovered in physics. The telephone company was not in that greater hurry to get it developed it into the field, after all have pretty much monopoly, why hurry? Now as you know, we are not willing to wait for scientific principles to develop we want the field tomorrow. So two fields to come together like that.

And the leisure which we use long ago, which we are still using some extent, develop the ideas first and then apply it is going to be less and less acceptable. When ideas first around you want to apply it. I just read last night, one of the presidents who were was at a museum or one of these World's Fair was shot, and the boat was back. Right his back, but the doctors refused operate because they didn't know where the bullet was. But there at the booth thing where X ray being demonstrated, they didn't use a new technique was right ready available, they kind of wheeled in their kind of picture. No, they were conservative, we don't

allow that much anymore, we're pushing very hard Are you going to be pushed very much to go from idea to develop item and get it on the market rapidly.

I once read there was some 76 different methods of predicting the future, which is why I'm engaged in doing to some extent. One is to predict tomorrow will be like today. Whatever temperature is today predict tomorrow is the same. It's a pretty good prediction. A somewhat better one is to note the linear trend and predict a linear trend. That's good for a while but not too long. And furthermore, it depends on which variable you pick to be linear. If you pick the coefficient from to be linear, one thing we pick the exponent something else. It doesn't work too well. I made many predictions on how much computing I'll do pretty soon. Because I need to know how much capacity would need and so on. I was regularly raw on the low side.

So one time I said I will predict high. So I got the form is and predicted real high a couple years later, the paper turned off my desk I looked at it, I was low again, the growth in computing has been unbelievable. On the other hand, on the other side, take artificial intelligence. The predictions made by almost all the experts 10, 20, 30 years ago have not been realized. So you can't always go on things. None the less, there's a saying. Short term predictions are optimistic long term predictions are pessimistic. And the reason is very simple. The long term of the pessimistic is nobody can believe in geometric regression. I say again, when we got the transistors going, nobody in his right mind would have predicted a million transistors on a ship that big. Nobody. It's beyond belief. But that's what we did. You know, so predict the future is a very, very hard business. But you have to do it. History is important.

Now, some people believe that history repeats itself, and some people believe exactly the opposite. But one thing you can be sure. What we now regard is the past was to some people the future. And what you think is the future will be the past. There'll be a time when some of you will be in the history books. Yes, you live long enough and do enough. And you end up in history books. So what do you think the future will become the past?

Now, another thing against history is Henry Ford Senior's remark history is bunk. And I think he said it for two reasons. One is history is rarely reported correctly. There are great main description of what happened at last almost during the war. No two of them agree. And they don't agree with what I think happened. Indeed, one time, a math teacher who wrote his experience about the matter. And publisher guy came in last almost our regular summer visit said to my friend, "I just read William's book. That is the how I remembered it." He said, "That isn't how I remember to either." I was just going to say, "How do you remember?" And I suddenly realized no two people remember the same. You're familiar with this an accident. Several witnesses see it, they report different things. There is no reliable report of what happened in the past. It's what's has come down to is accepted. Secondly, I think in affords mine was the fact that the past has been more rapidly disconnected from the future.

The invention of the computer tells you how much the world is different than what was before computers appear. It's a change in the way we do things. Engineering now is to great extent, getting a computer do job writing program and putting some terminal equipment around the defect the real world. The heart of much of engineering now is a computer. Now, some historians when you read them, they will give you the impression that the it was inevitable this was going to happen it was inevitable that Rome would fall or this or that. And on the other hand, they will tell you the future is very open ended. Many things are possible.

Can this be true that the past was very determined, the future is very open? It seems unlikely. So your left was saying maybe the past was not so determined. For example, individual lives of Alexander the Great, Napoleon, and Hitler. If they had died in their childhood, would not the world be very different?

On the intellectual side, Pythagoras, Aristotle, Newton, Maxwell, Einstein are examples who people who had they died in their youth, the world would be rather different. So individuals do matter. I suggest the past was less determine the historians like to make and the future is less open ended than you would like to believe. But there's a great many possibilities for you. The future had got great possibilities. Now, one of the thing is history is unforeseen technological inventions can ruin anything like I told you transistors, the development of vacuum tubes was practically cut off.

A technological invention, you can change completely, the history of something, and one could hardly foresee technological inventions. But they're also social inventions, which are important. You people have been trained mainly in the physical side, I've got to make you more sensitive to the fact that all of your life takes place in a social society, which has restraints. Thus I will claim that the future of technology will be less determined by what technology can do. Then social, legal, and other restraints on what we can do this, if you stop, think about highway controlled, computer control highway traffic. It sounds good, do you ask yourself who do I sue in an accident? And you begin to decide, you know, it's going to be a very, very difficult thing to get going. Very difficult. Social conventions are going to stop great things from happening.

Now I want to talk another thing, a story which I'll use several times the story of the drunken sailor. He staggered a couple steps this way, and he staggers this way. And he staggers this

way and he staggers this way. In N steps, typically he'll get the square root of N distance. In 100 steps, he'll get about 10. In 10,000 steps, it'll be about 100 times where he may be right where you started maybe for the way, but that's typical. On the other hand, if there's a pretty girl over there, he's talking like this back like this over like this. He's going to be disproportional the end. If I can create in you a vision of where you are headed, you will make a progress proportional to end. If you do not have a vision, you will wander like a drunken sailor, and get very little. So one of my major purposes is to get you to form a reasonable vision of what you are going to do your future what kind of a person you're going to be.

Now you're gonna say me, "But Hannick, how do I know the future?" I'm gonna say, "It doesn't matter much from our examine in life. What goal you set? What do you want march that way, that way or that way. If you have a goal, you'll get somewhere near it. If you don't have a goal, you're a drunken sailor." My problem is to make you form your goals and some except try to achieve them to make something important rather than just drifting. Now is comfortable drift to life. A great many people one question closely will assert that perfectly content to drift through life. I don't think too good idea of the whole thing.

Now, it's none of my business, what goal you take, it is my business, to force you one way or another to set up some reasonably decent goals to try and achieve something in your life.

Again, this society is paying a great deal of money for your education, It's entitled to something those who do something generally have some kind of goals to see where they're headed, and their lives add up. Those who don't are just a bunch of isolated events. They did this they did that they did nothing, but nothing added up. So I promise to get you to choose your goals. Even if you want Mary be a great guitar player, I don't mind. So long you set a

goal is struggling. That is the essential part that I'm really after. And that's what this course is about to some extent, forcing you somehow rather to do more than you would have done otherwise.

Now the standard method teaching is to have departments. Departments break things up into something better like calculus, linear, linear programming and so on. Too much falls between and this course is an attempt one way to plug all those holes, the engineering courses you had. You had a lot of engineering courses, they taught you this at the I mean, there are vast holes between them. The optimizing of the combos individual courses, is not optimizing a total education is I will come to the system engineering.

Now another goal I have is to show you that in spite of different departments, there's essential unity of all knowledge. When you face a difficult problem of unknown type, it doesn't matter whether it comes from chemistry, physics or anything else, you have to find the answer. And knowledge is pretty homogeneous, then it's no longer divided up into courses, no longer divided up in departments, although at Bell Labs, I was in the math department almost all the time. In fact, I was doing great many things. I was doing statistics I was doing computing, how you doing physics, I did a lot of other chemistry.

We did not observed tied to division but for purpose of organization, you do have to have some structure by want to get new minds. Now which is sort of a homogeneous body, which we have specialized with certain names, but it's all reconnected together. Now the course will center around computing. Not I like to think because I'm prejudiced my life in computing, but rather in fact they are going to dominate science and engineering. And there are reasons for this, very powerful reasons.

Economics, for example, computers are far cheaper than human beings. Far cheaper and getting cheaper by the year, humans are getting more expensive by the year. Speed. Far, far faster. Your nervous system if you drop something on your toe signals up your head about 100 meters per second. Like 1000 kilometers per second you walk in a league you can't even touch electronic speeds, there's no way you come near. So speed is overwhelmingly on the side the machine. Accuracy the me number ditches arithmetic carry. Yes, they can be quite precise. They can do double precision if necessary. You will have trouble doing double precision with take probably if you tried doing it. You can work it out but you'd have trouble. Reliability. They're far, far ahead of you, God or nature however you want to it, didn't make you to be a reliable thing. You've been walking for years and still every now and then you trip and stumble. You can't do anything really reliable. That's why man ended up at the top of the heap.

He has the flexibility built in. But don't ever try to get humans do something reliable. Take for example bowling. Why you just throw the ball down the alley exactly same way every time have a perfect game. Perfect games are rare even among the most skill experts.

Precision, flying and other things are very hard to do. We recognize it being very precise drill teams and so on or something remarkable. The human animal was really designed to do that. He was designed for something else. Repeated repetitive control because the machines got rapid control. We are now building airplanes which are basically unstable and we have a computer every millisecond is correcting usability.

So we get better performance out of it, but the pilot couldn't do it. If that computer goes out, the pilot's through. The pilot is left with a large scale abroad planning but the millisecond to millisecond is left better computer because of human just can't act act fast.

Another one who tried well on very much freedom boarding. It sounds trivial. You cannot put a human being on a job to look for something for three years and when it happens respond properly. You can put a computer on the job. You can put the computer on job to watch for the rare event. If such and such an episode happens in the atomic pile, do this. But that hasn't happened for four years. The human being isn't going to do very well. You guys know looking at think for last two and a half years even.

You can't get humans to be freed from boredom. Machines don't know what the word is.

Bandwidth in and out. In any rapidly changing situation, the person in charge can only get so much information in and out and there's a general belief that really you can process only about 50 bits per second maybe 60, something like that. But you can't process 10,000 bits per second. A machines got enormous more bandwidth. Now the visual auditory or pull all your inputs together they won't match a modern machine for bandwidth. Now only coming in, but getting orders out. For central control the humans simply cannot in a complicated situation compete with the machine. If it is merely bandwidth and bandwidth out. If it is making judgments to sell the story, but the machines simply cannot cope with us. We no longer have a crew aiming a gun at airplane, we have a self contained. The human is too slow, it just isn't much good.

We need much more rapid things and humans can cope with the bandwidth in and out, which is we speed of getting information is fundamental. Computers have got all over you. Ease of retraining. Training the old ways you learn to do something and now we I changed the equipment you gotta unlearn the old habits learn some new ones and you got to repeat them many many times to learn them with the computer I changed the program. And it's done, no elaborate training, no endless hours or constant practice. Just put a new program

and machine behaves a new way. Very easy. Hostile environments, outer space, underwater, high radiation fields, warfare manufacturing situations are unhealthy and saw how you put machines those situations burn humans are very very difficult. In space, I gotta keep this human being in atmosphere somewhat he's used to, oxygen. So I'll ask the employed high radiation will kill him and so on. How we're gonna manage to get people to Mars and back in the radiation field is coming from the sun, I don't know. Well, we'll sort of radium thoroughly or maybe decide not to send human beings that far. It's a problem.

Now personal problems with another man as well. It's one I'm much sensitive to. Personal problems dominate management. There are all kinds of trouble with people. With machines are no pensions. There are no personal squabbles, two machines don't get squabbling with other, but I've had two girls squabble who wouldn't even share the same room together. Unions, no. Personal leave, no. Eagles, no. Death of relatives, no. Your mother died, machines don't have that. Recreation. I turn the machine off that's the end of it. Human being, I have to provide reasonable recreation. Machines got all over humans. Now all of you probably already been saying, "Oh yeah, but what about the advantages humans have?" I will have to list those you're trying to do it already.

But I gave you a bunch of details, which you could find very hard to get around because the machine has got great advantage many places. And because it's economically sound, you are going to see more and more machines running organizations. Some computer, let's say computers, the design of chips is only computer controlled agree step. Some computers are actually being assembled heavily by machines. I was on the board of directors of a computer company for a while. And at one point, more than half the computers coming down the production line we were grabbing to mechanized a production. Were mechanized in the

building of computers. More than half the computers, we sold less than half of them because we're mechanized in line and getting production much cheaper. As show you how rapidly a company computing business was really mechanized itself. And one of my friends said he ordered a bunch of machines a message came in overnight, a bunch of machines assembled those particular computers they wanted. And the next day those computers were on the loading dock design, just what they wanted for the parts they want.

Now lastly, this is a certain sense of religious course, I am preaching the message that with one life to lead. You ought to me more than just get by. Now there are going to be religions. And I don't want to get involved in ones or the other too much. It is however, an emotional matter I'm really appealing to. Now is perfectly said that a happy like is one who has some goals they achieve. Well, starting the matter over and read about and talk to people, everybody pretty much up to agrees that it's not the achievement of the goal. That really is the best part is the struggle. The struggle to success is what makes you what you will be. Remember, you all age, you to live with yourself. There's no escaping live with yourself, your old age, you're stuck with yourself. And in old age you can't change much as you can when you're younger. Consider the kind of person you wish to be in your old age and start now being that kind of a person.

This is what the course is all about, really. In one sense. Now it's opinion. It's not a fact this opinion that most people believe that the struggle to to achieve excellence is worth the struggle. Also, when you look at people's lives, I can tell you a story which I may repeat a couple of times. As a child, I went to a movie. They were called Nickelodeons my day but we actually spent a dime to go to the movie. One Saturday, I went with a friend of mine, and it was one of these, you laughed and laughed and laughed. All ridiculous situations, we walked

out. And he said to me, "You know, that wasn't a very funny movie." I thought for a while. So you're right. All the laughter did not make a movie funny at all.

The same way with life. Pleasant life is not the ones the sum total of the pleasant moments. Somehow or others added up very, very differently. The Good Life is not the life of pleasure from moment to moment. And you know, the fact you are well aware that you cannot get up in the morning and say, I shall be happy today and make it work. The Good Life has to be snuck up upon. And I'm saying with an opinion of myself and many other books. The way to do that is to take yourself on hand and manage yourself to be the person you wish to be to achieve the goals you wish, and be more articulate than just idle drifting like a drunken sailor.

Now in ancient Greece, our boy Socrates said, "The unexamined life is not worth living." So what I'm saying goes back that far. I was crossing the campus one time as consultant for the present job. Hope for you put together I walk cross. I heard a professor walking across the campus right ahead of me saying to a student, "The unexamined life is not worth living." And in the course of crossing one quadrangle, he managed to say it three times. So I'll repeat the third time. "The unexamined life is not worth living."